

Soumyajit Karmakar

✉ skarmakar@ucsd.edu •  [SoumyajitKarmakar](https://github.com/soumyajitkarmakar) •  [in:soumyajit-karmakar](https://www.linkedin.com/in/soumyajit-karmakar)
 [soumyajitkarmakar.github.io](https://github.com/soumyajitkarmakar) •  Scholar

EDUCATION

University of California San Diego

2024 - 2026

Master of Science in Computer Science and Engineering.

Thesis Advisor: [Prof. Mikhail Belkin](#).

GPA: 4.0/4.0

Indian Institute of Information Technology, Guwahati

2019 - 2023

Bachelor of Technology in Computer Science and Engineering.

CGPA: 9.35/10.00 (Silver medalist)

RESEARCH EXPERIENCE

Indian Institute of Science, Bengaluru

Vision and AI Lab (VAL)

Project Assistant - Advisor : [Prof. R. Venkatesh Babu](#)

September 2023 - August 2024

- Collaborated with Fujitsu Research and developed CLIP-based image retrieval for person re-identification.

International Institute of Information Technology, Hyderabad

Center for Visual Information Technology (CVIT)

Research Fellow - Advisor : [Prof. C. V. Jawahar](#)

January 2023 - August 2023

- Worked on developing Diffusion Models for semantic segmentation in the urban road settings.

University of North Carolina at Charlotte, North Carolina

Research Intern - Advisor : [Dr. Srijan Das](#) and [Dr. Michael S. Ryoo](#)

August 2022 - March 2023

- Developed a joint training framework using a Self-Supervised Auxiliary Task to enhance the performance of ViTs on small datasets.

CSIR-CEERI Pilani, Rajasthan

Advanced Information Technologies Group (AITG)

Research Intern - Advisor : [Dr. Sanjay Singh](#)

May 2022 - July 2022

- Developed a novel few-shot learning framework, using a Convolution based ensembling technique, for anomaly detection.

ACHIEVEMENTS

- **Global Rank 1**, in the Parameterized Algorithms and Computational Experiments (PACE) 2022 (Heuristic Track). [Link](#) for the paper. [Link](#) for the solver.

PUBLICATIONS

- **Soumyajit Karmakar***, Shrinivas Ramasubramanian*, Harsh Rangwani*, Yasuhiro Aoki, Genta Suzuki, Venkatesh Babu. "Prompt-ReID: Prompt Based Inference for Robust Re-Identification". (*Pre-print*)
- Srijan Das, Tanmay Jain, Dominick Reilly, Pranav Balaji, **Soumyajit Karmakar**, Shyam Marjit, Xiang Li, Abhijit Das, Michael Ryoo. "Limited Data, Unlimited Potential: A Study on ViTs Augmented by Masked Autoencoders". In Proceedings of the 2024 IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**) 2024, Waikoloa, Hawaii, USA.
- **Soumyajit Karmakar**, Abeer Banerjee, Prashant Sadashiv Gidde, Sumeet Saurav, Sanjay Singh. "Convolutional Ensembling based Few-Shot Defect Detection Technique". In Proceedings of the 2022 Indian Conference on Computer Vision, Graphics and Image Processing (**ICVGIP**), IIT Gandhinagar, India.
- Aman Jain, Sachin Agarwal, Nimish Agrawal, **Soumyajit Karmakar**, Srinibas Swain. "Feedback vertex set using Edge Density and REmove Redundant (FEDRER): A heuristic solver for finding a feedback vertex set in a directed graph". In poster session of the 2022 International Symposium on Parameterized and Exact Computation (**IPEC**), Potsdam, Germany.